Computer Science - Undergraduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following decimal numbers has an exact representation in binary notation?
   - (a) 0.1
   - (b) 0.2
   - (c) 0.3
   - (d) 0.5
2. Which of the following language features requires that stack-based storage allocation be used rather than static allocation?
   - (a) Recursive procedures
   - (b) Arbitrary goto's
   - (c) Two-dimensional arrays
   - (d) Integer-valued functions
3. A 3-way, set-associative cache is
   - (a) one in which each main memory word can be stored at any of 3 cache locations
   - (b) effective only if 3 or fewer processes are running alternately on the processor
   - (c) possible only with write-back
   - (d) faster to access than a direct-mapped cache
4. Church's thesis equates the concept of "computable function" with those functions computable b
   thesis?
   - (a) It was first proven by Alan Turing.
   - (b) It has not yet been proven, but finding a proof is a subject of active research.
   - (c) It can never be proven.
   - (d) It is now in doubt because of the advent of parallel computers.
5. Of the following sorting algorithms, which has a running time that is LEAST dependent on the initial ordering of the input?
   - (a) Insertion sort
   - (b) Quicksort
   - (c) Merge sort
   - (d) Selection sort

True / False (10 marks)

Answer True or False

6. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes i
7. True or False: The addition of the two's-complement numbers 10000001 and 10101010 results i
8. True or False: Local caching of files in distributed file systems always results in increased networ
9. True or False: The one-time pad is considered a perfectly secure encryption scheme when used
10. True or False: The access matrix approach, if stored directly, can be large and cumbersome to m

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to iterate over elements repeating each as many times as its count.
12. Write a python function to find minimum sum of factors of a given number.

End of Paper